HW 2 DUE 2/15

1) a) AB is symmetric $\iff AB = BA$

($\Rightarrow$) suppose AB is symmetric, $AB = (AB)^T$.

$$AB = (AB)^T = B^T A^T = BA$$

($B^T = B$, $A^T = A$)

so $AB = BA$.

($\Leftarrow$) Suppose $AB = BA$.

$$AB = BA = B^T A^T = (AB)^T = AB$$

(since A + B are symmetric)

so AB is symmetric.

b) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ A and B are symmetric.

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$AB \neq AB^T$, so AB is not symmetric.

2) a) $\det\begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 2 & 2 & 5 \end{bmatrix}$ = Choosing row 1. $\det A_{11} = \det\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} = 9$

$$\left[(-1)^{1+1}\cdot 1 \cdot 9\right] + \left[(-1)^{1+2}\cdot 1 \cdot (-1)\right] + \left[(-1)^{1+3}\cdot 1 \cdot (-4)\right] = \det A_{12} = \det\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} = -1$$

$$9 + 1 - 4 = \boxed{6} \qquad \det A_{13} = \det\begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} = -4$$

b) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$ $\xrightarrow[r_2 \leftrightarrow r_3]{r_1 \leftrightarrow r_4}$ $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

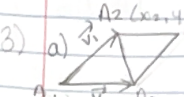
A B

Since this is upper triangular, can do $1 \cdot 1 \cdot 1 \cdot 1$. $\det(B) = 1$.

Since we did 2 row swaps, $\det(B) = -(-\det(A))$

$\det(B) = \det(A)$

$k = 2$, and since k is even, $\det(A) = \det(B) = \boxed{1}$

3) a) 

① a c
$\vec{v}_1 = (x_2 - x_1, y_2 - y_1)$
$\vec{v}_2 = (x_3 - x_1, y_3 - y_1)$
b d

② $|\det \begin{bmatrix} a & b \\ c & d \end{bmatrix}|$

$\frac{1}{2}\left|\det \begin{bmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{bmatrix}\right| =$

③ $\frac{1}{2}\left|\det \begin{bmatrix} x_2 & x_3 \\ y_2 & y_3 \end{bmatrix}\right| = \boxed{\frac{1}{2}|x_2 y_3 - x_3 y_2|}$

④ $\frac{1}{2}\left|\det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}\right|$ expand on column 1

$\det A_{11} = x_2 y_3 - y_2 x_3$
$\det A_{21} = x_1 y_3 - y_1 x_3$
$\det A_{31} = x_1 y_2 - y_1 x_2$

$\frac{1}{2} \cdot \left[(-1)^{1+1}(1)(x_2 y_3 - y_2 x_3)\right] + \left[(-1)^{2+1}(1)(x_1 y_3 - y_1 x_3)\right] +$

$\left[(-1)^{3+1}(1)(x_1 y_2 - y_1 x_2)\right] = \frac{1}{2}\left[x_2 y_3 - y_2 x_3 \underbrace{- x_1 y_3}_{0} + \underbrace{y_1 x_3}_{0} + \underbrace{x_1 y_2}_{0} - \underbrace{y_1 x_2}_{0}\right] =$

$= \boxed{\frac{1}{2}|x_2 y_3 - x_3 y_2|}$ Since both equations circled evaluate to equal values, we have proven the formula.

4) a) $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $0+0+0 \neq 3$ Since $\vec{0}$ is not in W, W is not a subspace of $\mathbb{R}^3$

b) $\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ $9+16=25$ ✓ $\vec{v}$ $\quad \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ $0+4=4$ ✓ $\vec{w}$ $\quad \begin{bmatrix} 3 \\ 6 \\ 7 \end{bmatrix}$ $9+36 \neq 49$ $45 \neq 49$ $\vec{v}+\vec{w}$ not in W

$\vec{v}, \vec{w}$ in W such that $\vec{v}+\vec{w}$ not in W so these sets are not a subspace of $\mathbb{R}^3$.

6) a) column: $\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \\ 1 & 4 & 2 \end{bmatrix} \xrightarrow[r_4 - r_3]{r_2 - r_1} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_3 - r_1} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow[r_3(\frac{1}{2})]{r_1 - r_2}$

$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow[r_4 - r_2]{r_3 - r_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Third column has no leading one, so $\vec{v}_3 = -2\vec{v}_1 + \vec{v}_2$

Linearly dependent

row: $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ -1 & 0 & 1 & 2 \end{bmatrix} \xrightarrow[r_3 + r_1]{r_2 - r_1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} \xrightarrow[r_3 - r_2]{r_1 - r_2} \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Zero row so again, linearly dependent.

7) a) $A = \begin{bmatrix} 1 & 5 & 2 & 0 & 0 \\ 1 & 11 & 5 & 1 & 0 \\ 0 & 6 & 3 & 1 & 0 \end{bmatrix} \xrightarrow{r_2 - r_1} \begin{bmatrix} 1 & 5 & 2 & 0 & 0 \\ 0 & 6 & 3 & 1 & 0 \\ 0 & 6 & 3 & 1 & 0 \end{bmatrix} \xrightarrow{r_2 - r_3} \begin{bmatrix} 1 & 5 & 2 & 0 & 0 \\ 0 & 6 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_2(\frac{1}{6})}$

$\begin{bmatrix} 1 & 5 & 2 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1 - r_2(5)} \begin{bmatrix} 1 & 0 & -\frac{1}{2} & -\frac{5}{6} & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$x_1 = -\frac{1}{2}x_3 - \frac{5}{6}x_4$

$x_2 = \frac{1}{2}x_3 + \frac{1}{6}x_4$

x_3 is free

x_4 is free

vec form: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_3 - \frac{5}{6}x_4 \\ \frac{1}{2}x_3 + \frac{1}{6}x_4 \\ x_3 \\ x_4 \end{bmatrix}$

LC form: $x_3 \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{5}{6} \\ \frac{1}{6} \\ 0 \\ 1 \end{bmatrix}$

basis of ker(A) is: $\left\{ \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/6 \\ 1/6 \\ 0 \\ 1 \end{bmatrix} \right\}$

basis of image of A: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 11 \\ 6 \end{bmatrix} \right\}$

3) b) $\frac{1}{2}\left|\det\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}\right|$

$\det A_{11} = -1 - 0 = -1$

$\det A_{12} = -1$

$\det A_{13} = -1 - (1) = -2$

$\frac{1}{2}\left|\left[(-1)^{1+1}(1)(-1) + (-1)^{1+2}(0)(-1) + (-1)^{1+3}(1)(-2)\right]\right| =$

$\frac{1}{2}\left|-1+0-2\right| = \boxed{-3/2}$